

Hence p is proportional to $V^{1/2}$ and λ is proportional to $V^{-1/2}$

30 A

$$p = mv = (9.11 \times 10^{-31})(1.50 \times 10^6) = 1.37 \times 10^{-24} \text{ kg m s}^{-1}$$

$$\Delta p = 0.002 p = 0.002(1.37 \times 10^{-24}) = 2.74 \times 10^{-27} \text{ kg m s}^{-1}$$

$$\Delta x = h / (2.74 \times 10^{-27}) = 2.41 \times 10^{-7} \text{ m}$$